Fe-C Problem 3 | Problems T4_PhaseDiagrams II.pdf #3 | Fe-C · hypereutectoid steel

Phases and Microconstituents of a 1.5 % C Steel

Original statement · Problems T4_PhaseDiagrams II.pdf, p. 2

3.- For an alloy Fe-1.5%C, at the following temperatures: T_1 = (1147+ΔT)ºC; T_2 = (1147-ΔT)ºC; T_3 = 900ºC; T_4 = (727+ΔT)ºC; T_5 = (727-ΔT)ºC. a) Indicate the phases present, their chemical composition and their mass fraction. b) Indicate the micro constituents, their chemical composition and their mass fraction at T_5.

Official Solution: a) T_1 **and** T_2**:** C_γ **= 1.5% C,** w_γ **= 1;** T_3**:** C_{Fe_3C} **= 6.7% C,** C_γ **= 1.3%** **C,** w_γ **= 0.96,** w_{Fe_3C} **= 0.04;** T_4**:** C_γ **= 0.76% C,** C_{Fe_3C} **= 6.7% C,** w_γ **= 0.87,** w_{Fe_3C} **= 0.13;** T_5**:** C_α **= 0.022% C,** C_{Fe_3C} **= 6.7% C,** w_α **= 0.78,** w_{Fe_3C} **= 0.22; b) Pearlite** $w_{Fe_3C'}$ **= 0.13,** $w_{pearlite}$ **= 0.87,** $w_{\alpha E}$ **= 0.78,** $w_{Fe_3C,E}$ **= 0.09**

PHASE 1 · HYPOTHESES & PARAMETERS

1. Equilibrium cooling, Fe-Fe3C diagram [Session 9 Slide 4].
2. $0.76 < C_0 = 1.5 < 2.14$: a **hypereutectoid steel**; at 1147 ± ΔT it is single-phase austenite (1.5 < 2.14).
3. **Reading of the figure:** at 900 °C the A_{cm} line is at about 1.3 % C (the key's value; a straight line from (0.76, 727) to (2.14, 1147) would give 1.33).

PHASE 2 · FRAMES & STRATEGY

Frame: the Fe-Fe3C diagram (wt% C, °C); one tie line per temperature. Below A_{cm} proeutectoid cementite precipitates from the austenite [Session 9 Slide 19]; at 727 °C the remaining austenite (0.76 % C) becomes pearlite.

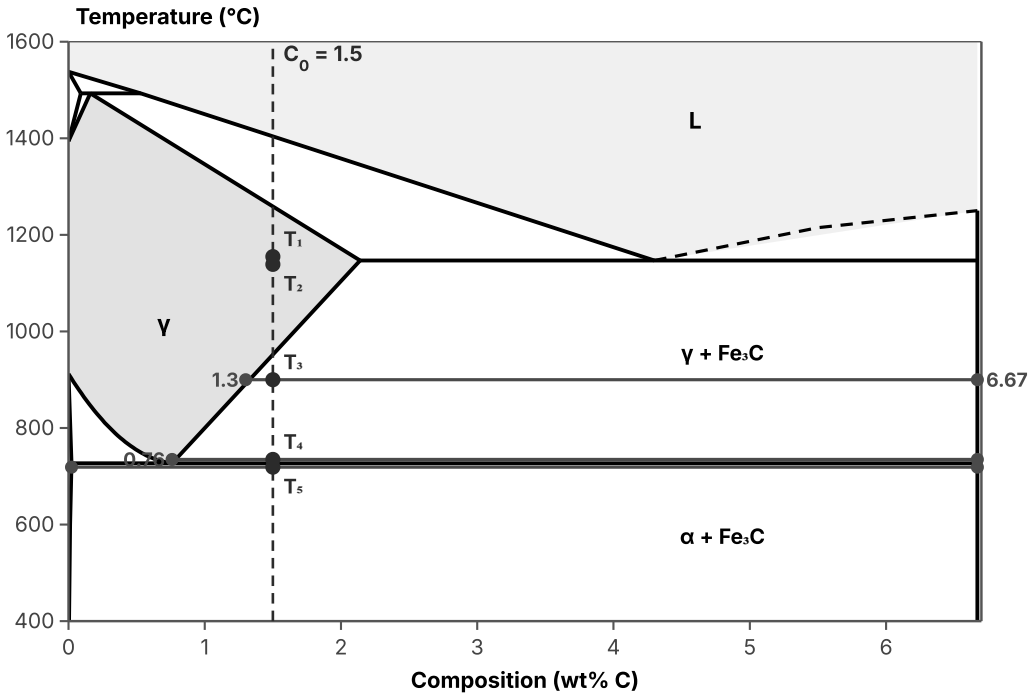


Figure 31. Fe-1.5 % C: γ at 1147 ± ΔT; γ + Fe_3C at 900 °C (γ of about 1.3 % C, a reading) and just above 727 °C; α + Fe_3C below 727 °C.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases T_1, T_2: γ, 1.5 % C, $W_\gamma = 1$.

T_3 = **900 °C** (γ 1.3 / Fe_3C 6.67; length 5.37):

$$W_\gamma = \frac{6.67 - 1.5}{6.67 - 1.3} = \frac{5.17}{5.37} = 0.963 \qquad W_{Fe_3C} = \frac{1.5 - 1.3}{5.37} = \frac{0.2}{5.37} = 0.037$$

T_4 (γ 0.76 / Fe_3C 6.67; length 5.91):

$$W_\gamma = \frac{6.67 - 1.5}{5.91} = \frac{5.17}{5.91} = 0.875 \qquad W_{Fe_3C} = \frac{1.5 - 0.76}{5.91} = \frac{0.74}{5.91} = 0.125$$

T_5 (α 0.022 / Fe_3C 6.67; length 6.648):

$$W_\alpha = \frac{6.67 - 1.5}{6.648} = \frac{5.17}{6.648} = 0.778 \qquad W_{\mathrm{Fe_3C}} = \frac{1.5 - 0.022}{6.648} = \frac{1.478}{6.648} = 0.222$$

b) Microconstituents at T_5 Proeutectoid cementite = the cementite at T_4, $W_{\mathrm{Fe_3C'}} = 0.125$; pearlite = the austenite at T_4, $W_{\mathrm{pearlite}} = 0.875$. Inside the pearlite: $W_{\alpha,E} = 0.778$ (all the ferrite) and $W_{\mathrm{Fe_3C},E} = 0.222 - 0.125 = 0.097$.

PHASE 4 · CHECKS & INTERPRETATION

- $0.778 + 0.097 = 0.875 = W_{\mathrm{pearlite}}$; and $0.875 \times 0.889 = 0.778$.
- Mass balance at T_5: $0.778 \times 0.022 + 0.222 \times 6.67 = 0.017 + 1.481 = 1.50\ \%\ \mathrm{C}$.
- The proeutectoid cementite grows from 0.037 at 900 °C to 0.125 at 727 °C as the austenite loses carbon along A_{cm}.

Check against the official solution

Agrees with the key, except for the last figure.

DISCREPANCY WITH THE OFFICIAL SOLUTION

The key gives $w_{Fe_3C,E} = 0.09$. Exactly, $0.222 - 0.125 = 0.097$ (≈ 0.10), confirmed by $0.875 \times 0.111 = 0.097$; the key truncates the difference of the rounded values $0.22 - 0.13$. The computed value is kept.